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Preserving the Cultural Heritage of Indonesian Society through Digital Preservation in Libraries

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Abstract

Digital preservation plays a crucial role in safeguarding Indonesia's cultural heritage by ensuring the longevity of historical documents, artifacts, and traditional knowledge. However, efforts in digitalization face significant challenges, including technical infrastructure limitations, legal ambiguities, and public engagement barriers. This study examines the role of libraries in digital preservation, identifying key benefits, challenges, and potential strategies for sustainable implementation. Through a thematic analysis of 16 relevant studies selected from a comprehensive review of 500 academic articles, findings indicate that digitalization enhances accessibility and efficiency but is hampered by financial constraints, inadequate technological resources, and insufficient legal frameworks. Furthermore, the lack of skilled professionals in digital archiving exacerbates implementation difficulties. The study highlights the need for stronger government policies, increased funding, and collaborative efforts between libraries, policymakers, and cultural institutions. To strengthen digital preservation strategies, the study recommends integrating advanced technologies such as artificial intelligence and blockchain, ensuring long-term sustainability. Additionally, hybrid approaches that merge digital and physical interactions can enhance public engagement with cultural heritage. This research offers a valuable reference for stakeholders in cultural preservation and digital library management, contributing to the broader discourse on digital heritage conservation.

Keywords

Digital preservation, cultural heritage, library digitalization, legal frameworks, technological challenges.



Introduction

Indonesia has a diverse cultural wealth, ranging from ancient manuscripts and traditional texts to various forms of art and customs passed down from generation to generation. This cultural heritage is a national identity that must be preserved so that it does not become extinct or forgotten by future generations (Wagner & de Clippele, 2023). However, the main challenges in maintaining this cultural heritage are the risk of physical degradation, loss of historical documentation, and lack of accessibility for the wider community (Bansode & Shinde, 2019; Gorman, 2007). In the digital era, libraries play a strategic role in cultural heritage preservation efforts through digitization, which allows information to be accessed and passed on in the long term (Anna, 2017; Barbuti, 2021; Immaculata & Ike, 2023).

Several previous studies have highlighted the importance of digitization in cultural heritage preservation (Purnomo & Pratiwi, 2023). Research by Zhou et al. (2012) showed that digitizing library collections helps preserve physical documents and increases accessibility for global users. In addition, another study by Manžuch et al. (2005) revealed that adopting digital technology in libraries can extend the life of cultural documents by preventing damage due to environmental factors (Bhatt et al., 2015). However, some studies are still limited in examining the digitization system's sustainability and implementation challenges, such as limited resources, technological infrastructure, and copyright protection policies (Khan et al., 2018; Ross et al., 2015; Shafi et al., 2014).

The limitations in previous research indicate the need for a digital preservation model that is more effective, sustainable, and suitable for the conditions of libraries in Indonesia. This research aims to fill this gap by exploring strategies to implement digital preservation in libraries to preserve the cultural heritage of Indonesian society. It also seeks to identify the main challenges faced in the digitization process and the solutions that can be applied to overcome these obstacles.

This research provides a new contribution in the form of a holistic approach to digital preservation in libraries, considering technical, policy, and social aspects. This study is expected to provide deeper insights into best practices in digitizing cultural heritage so that it can serve as a reference for libraries, governments, and other stakeholders. Thus, the results of this study are expected to contribute to developing more effective and sustainable digital preservation policies in Indonesia.

Methods

The qualitative approach used in this study to explore the role of libraries in the digital preservation of cultural heritage in Indonesia is highly relevant to developments in cultural heritage management. Literature studies show that digital preservation has become essential in safeguarding tangible and intangible cultural heritage from loss of meaning and valuable information (Isa et al., 2018; Loebbecke &

Thaller, 2011). For example, research by Isa et al. (2018). emphasizes the importance of digitizing intangible cultural heritage so that various cultural elements do not disappear. Similarly, Lukman et al. (2020) showed that the right approach to digitizing cultural heritage, as done in the Depok Lama project, greatly supports preservation efforts. Thus, as information storage institutions, libraries have a great responsibility in this process.

The research procedure is carried out in several stages, including literature searches using academic databases and digital library repositories, including specific keywords relevant to the research topic. Digital documentation, including 3D technology, is one method that is increasingly being applied in preserving cultural heritage sites, as outlined in a study by Crisnapati et al. (2019). Furthermore, analyzing the challenges faced in digital preservation, such as technical constraints, is important, especially to understand how technology can be optimized to support more effective preservation (Selmanović et al., 2020). The research procedure was conducted in several stages. First, a literature search was performed using academic databases and digital library repositories to obtain relevant references with specific keywords such as "digital preservation," "cultural heritage," and "digital library." Second, the literature was selected based on the research's relevance, quality, and novelty. Third, the data obtained was analyzed to identify the digital preservation strategies implemented in the library, the challenges faced, and the impact of digitization on the preservation of cultural heritage.

In the data analysis, using thematic methods to group information by categories, such as the benefits of digitization and socio-cultural impacts, provided more profound insights into best practices. This aligns with research from Cramer & Kott (2010), which discusses creating compelling digital preservation services. This research is expected to provide valuable recommendations to improve the effectiveness of cultural heritage preservation through digital libraries in Indonesia, thus aligning with strategies to expand access to and community engagement in cultural heritage (Constantinidis, 2016; Oomen & Aroyo, 2011). The data analysis used a thematic method to group information based on specific categories, such as the benefits of digitization, technical constraints, legal aspects, and social and cultural impacts. This approach allows the research to describe various aspects of digital preservation comprehensively and systematically.

The initial data was obtained using the Publish or Perish application by searching for keywords related to *digital preservation*, *cultural heritage*, and *digital libraries*. This search resulted in 500 articles. The next step involved filtering these results by journal and book types, leading to the identification of six books. As a result, 494 journal articles were selected for further analysis using thematic analysis. Another round of selection was carried out by eliminating articles that did not align with the study's theme, ultimately leaving 16 articles for the final thematic analysis.

Therefore, through this systematic approach, the research will provide an in-depth understanding of the role of libraries in digital preservation and contribute to

the development of better policies and practices for cultural heritage preservation in Indonesia.

Results and Discussion

Result

This study aims to analyze the role of libraries in the digital preservation of Indonesia's cultural heritage. The thematic analysis method identifies, categorizes, and interprets patterns emerging from various literature sources. This approach enables researchers to comprehend multiple aspects, including the benefits of digitalization, technical challenges, legal and regulatory elements, and the social and cultural impacts of implementing digital technology in heritage preservation.

I. Data Familiarization

The first step in the thematic analysis process is data familiarization, which involves reading and understanding all collected literature. The initial data was retrieved from Publish or Perish using *digital preservation*, *cultural heritage*, and *digital libraries*. The search yielded 500 articles, which were then filtered to distinguish between journal articles and books. This selection identified six books, leaving 494 journal articles for further analysis. The selection was further refined by eliminating articles not aligning with the research themes, ultimately resulting in 16 articles for thematic analysis.

At this stage, researchers reviewed the abstracts of the selected articles to gain an initial understanding of their content and scope. This familiarization process helped identify recurring concepts and understand how libraries leverage digital technology for cultural preservation.

II. Initial Coding

Once familiar with the collected data, the next step is initial coding. This phase involves identifying key terms, phrases, and primary concepts frequently appearing in the analyzed articles' abstracts. Word frequency analysis was utilized to detect patterns within the data.

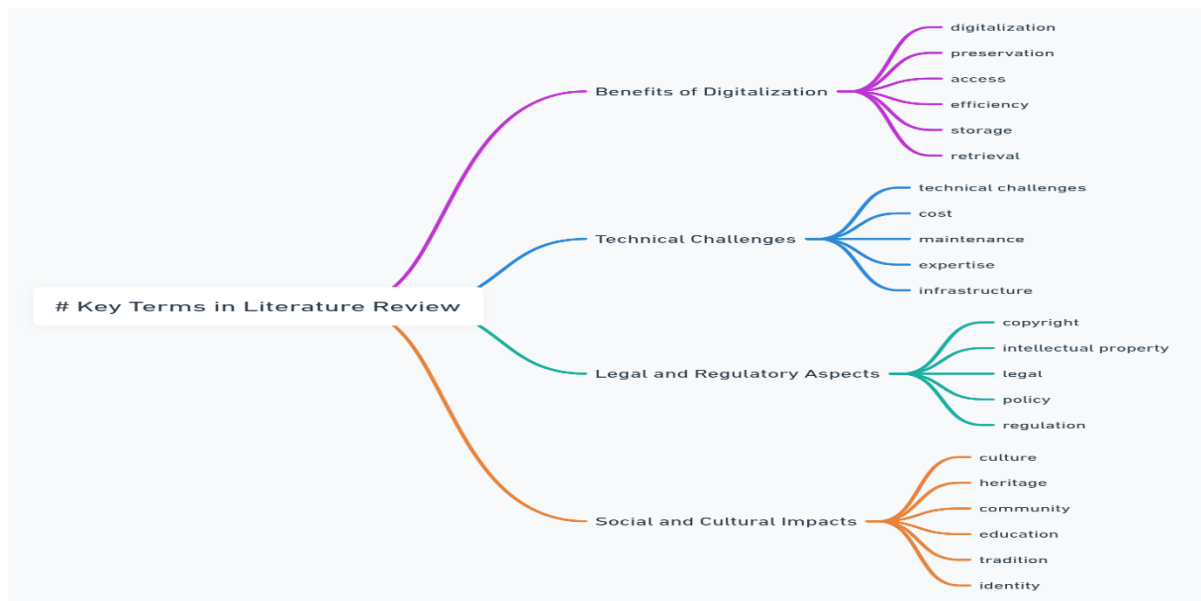


Figure 1. Key terms in literature review

Source: Author (2025)

Some key terms frequently found in the literature include:

1. Benefits of Digitalization: *digitalization, preservation, access, efficiency, storage, retrieval.*
2. Technical Challenges: *technical challenges, cost, maintenance, expertise, and infrastructure.*
3. Legal and Regulatory Aspects: *copyright, intellectual property, legal, policy, regulation.*
4. Social and Cultural Impacts: *culture, heritage, community, education, tradition, identity.*

These keywords form the foundation for the initial coding process, categorizing findings into four main themes.

III. Theme Identification

In this stage, the identified codes were grouped into overarching themes representing patterns within the data. After thorough analysis and categorization, four primary themes emerged in this study:

1. Benefits of Digitalization: Encompassing how digitalization enhances information accessibility, preserves physical collections from deterioration, and improves archive and document management efficiency.
2. Technical Challenges: These include the obstacles to implementing digitalization, such as high costs, limited digital infrastructure in particular institutions, and a shortage of skilled professionals in information technology.

- 3. Legal and Regulatory Aspects: Emphasizing the importance of copyright protection for digital collections and the need for clear open-access policies to balance legal protection and public access to cultural information.
- 4. Social and Cultural Impacts: Demonstrating that library digitalization fosters public appreciation of cultural heritage and strengthens national identity while potentially altering how society interacts with cultural resources.

Each theme is closely related to the research context. While digitalization offers new opportunities for cultural preservation, technical and legal challenges must be addressed to optimize its social and cultural impact.

IV. Theme Review and Refinement

The next step is theme review and refinement, ensuring that each identified theme accurately and comprehensively represents the data. This review involves comparing findings with established theories and previous research to confirm that no crucial information is missing or redundant in any category.

Evaluation results indicate that all themes are well-represented, requiring no significant revisions or deletions. However, some thematic aspects were refined for greater specificity, particularly in addressing technical challenges libraries face in digitalization, such as high infrastructure costs and the shortage of competent IT personnel.

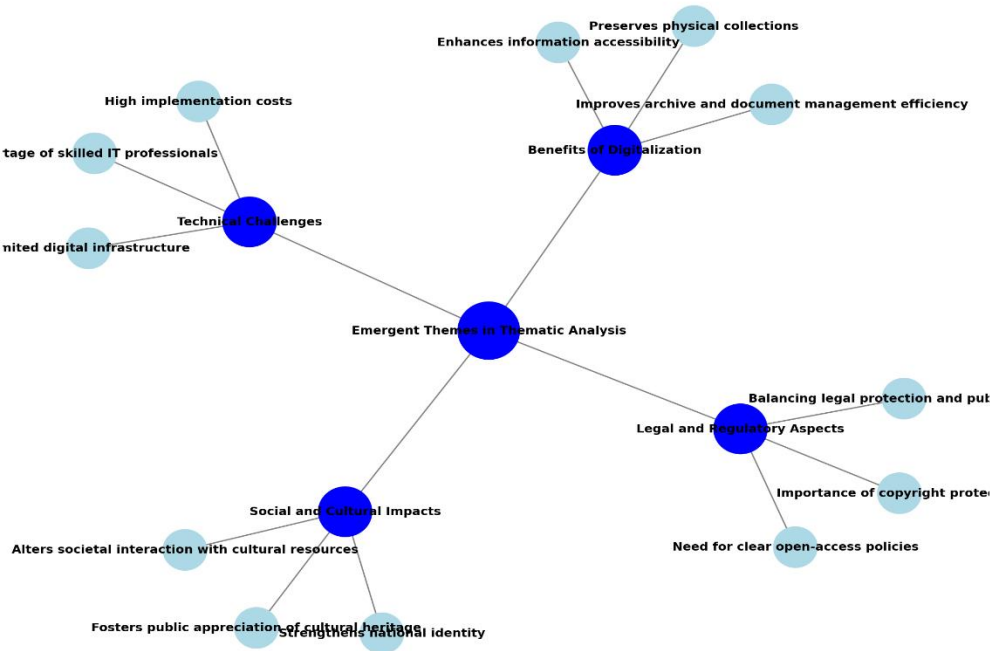


Figure 2. Emergent themes in thematic analysis
Source: Author (2025)

V. Interpretation and Report Writing

The final stage of the thematic analysis is interpretation and report writing, where each theme is further analyzed to provide deeper insights. The study confirms that the Benefits of Digitalization primarily include accessibility, physical document protection, and efficiency in collection management. However, technical challenges pose significant barriers, particularly in high equipment costs, limited infrastructure in regional libraries, and a lack of skilled personnel capable of managing digital systems. Regarding Legal and Regulatory Aspects, findings highlight the need for more apparent copyright and licensing policies, as libraries often struggle to share digital content without infringing intellectual property laws. Meanwhile, the Social and Cultural Impacts analysis reveals that digitalization has created new opportunities for the public to engage with and appreciate cultural heritage. However, challenges remain in ensuring that communities continue actively engaging in physical and digital cultural preservation.

Overall, this study underscores that digital preservation in libraries plays a crucial role in safeguarding Indonesia's cultural heritage. However, achieving optimal success requires a comprehensive strategy, including regulatory reinforcement, increased investment in technological infrastructure, and expanded digital literacy programs to support digital heritage preservation. Thus, this research serves as a foundation for further policymaking to strengthen the role of libraries in cultural preservation through digital technology.

The thematic analysis results demonstrate that digitalization is vital in cultural heritage preservation, particularly in enhancing accessibility, improving management efficiency, and protecting physical collections. However, technical and legal barriers remain significant challenges that must be addressed to ensure the sustainability of digitalization efforts. Therefore, collaboration among libraries, the government, and society is essential to maximize the benefits of digitalization while mitigating its challenges.

This study contributes to understanding how libraries can adopt digital technology for cultural preservation. The recommendations include strengthening digitalization policies, enhancing human resource capacity in digital technology, and developing more efficient systems to support large-scale digital preservation.

Discussion

I. The Role of Digitalization in Preserving Cultural Heritage in Indonesia

Digital preservation is crucial in safeguarding cultural heritage, ensuring that traditional knowledge, artifacts, and historical records remain accessible for future generations. In Indonesia, digitalization has become increasingly important as many cultural assets, including manuscripts, traditional music, and historical sites, face

threats from environmental degradation, natural disasters, and neglect. The findings of this study highlight the benefits, challenges, and social implications of digital preservation, emphasizing the importance of well-planned strategies and regulatory frameworks to support digital initiatives in libraries.

II. Benefits of Digitalization in Indonesian Libraries

One of the key findings of this research is that digitalization enhances accessibility, efficiency, and security in managing cultural heritage. Indonesian institutions, such as the National Library of Indonesia, have initiated large-scale digital preservation projects to archive rare manuscripts and historical documents (Shah et al., 2021). For example, the digitalization of *Negarakertagama*, a Javanese historical manuscript from the 14th century, has provided researchers and the general public with broader access to Indonesia's past (Pigeaud & Nijhoff, 1960). This aligns with findings in other countries, where digitalization has been proven to increase access to rare collections (Beagrie, 2004).

Furthermore, digital repositories have allowed Indonesian scholars to collaborate internationally by providing digital access to historical archives. The availability of digital collections has also fostered greater academic engagement and participation in global discussions on heritage conservation (Adetunji & Okuonghae, 2022). Universities such as Gadjah Mada University and the University of Indonesia have actively supported digital archive programs, ensuring that historical texts and cultural artifacts are preserved and widely disseminated (Rahmawati, 2022).

Digital preservation in libraries plays a pivotal role in safeguarding Indonesia's cultural heritage. However, this process presents opportunities and challenges that must be carefully considered to ensure long-term sustainability (Walia et al., 2023). The benefits of digital preservation include enhanced accessibility, protection of fragile collections, efficient content management, institutional collaboration, and improved restoration and conservation efforts (Kirchhoff, 2008). At the same time, digital preservation introduces technological dependency, potential loss of quality, security risks, limitations in digitizing certain materials, and a diminished physical experience in exploring collections (Friday & Eze, 2022). Libraries must navigate these complexities by adopting strategic technological solutions while ensuring that the cultural integrity of digitalized materials is maintained (Aisha & Dr.Baba, 2021).

One of the most significant advantages of digital preservation is increased accessibility. The availability of digital collections enables users to access valuable cultural heritage materials remotely, breaking geographical and temporal barriers. The National Library of Indonesia has pioneered this initiative through platforms like iPusnas, which provides open access to thousands of digital books and manuscripts. Similarly, the Yogyakarta Library's digitalization of Javanese manuscripts has facilitated global access to rare documents previously restricted to physical archives (Lukman et al., 2020). This advancement democratizes knowledge, ensuring cultural

heritage is available to a broader audience, including researchers, educators, and the general public.

Digital preservation also contributes to the protection and conservation of fragile materials. Many historical documents risk deterioration due to aging, environmental factors, or natural disasters. By converting them into digital formats, libraries ensure their longevity while minimizing physical degradation. Crisnapati et al. (2019) highlights that preservation strategies should prioritize preventing damage to original documents while ensuring their usability. This approach has been adopted by the Borobudur Conservation Office, which has digitally archived temple reliefs to protect them from weathering and pollution (Lintangkawuryan & Adiati, 2017).

Another key advantage of digital preservation is its efficiency in content management. Digital collections can be indexed, searched, and retrieved faster than physical archives, enabling users to locate relevant materials quickly. Libraries that implement metadata standards, such as Dublin Core and MARC 21, improve the discoverability of their digital assets (Formenton & Gracioso, 2020). This capability enhances research productivity, as users no longer need to sift through physical collections to find necessary materials manually.

Digital preservation also fosters collaboration between institutions, allowing libraries, museums, and archives to share collections across different platforms. This creates a broader network of cultural preservation efforts. The Indonesian Digital Library Network (IDLN) exemplifies this model, where multiple institutions contribute to a shared repository, enriching the available resources for academic and cultural research (Rizkia et al., 2024)

III. Challenges of digital preservation

However, despite these advantages, digital preservation also has some challenges. One of the main concerns is technology dependency. Reliance on digital systems means that technical failures, software obsolescence, or data corruption can lead to the loss of valuable information. Crisnapati et al. (2019) warns that inadequate infrastructure and outdated technology can make digital collections inaccessible. To mitigate this risk, libraries should implement a robust digital preservation framework, including regular data migration, cloud-based storage solutions, and redundant backup systems (Wulansari, 2017).

Another concern is the potential loss of quality in digital conversion. Some historical artifacts contain intricate details that may not be accurately captured in digital formats. Textures, embossing marks, and physical characteristics that contribute to the authenticity of a manuscript may be lost during scanning. Friday & Eze (2022) emphasize that while high-resolution scanning can improve preservation accuracy, some aspects of the authenticity of physical objects cannot be replicated digitally. This limitation calls for a hybrid preservation strategy where digital archives complement, rather than replace, physical preservation efforts.

Security risks are another key issue in digital preservation. Digital collections are vulnerable to cyber threats, including hacking, unauthorized access, and data breaches. The increasing frequency of cyberattacks targeting library databases highlights the need for stringent cybersecurity measures (Rinehart et al., 2014). Encryption, access control mechanisms, and regular security audits are essential to protect digital assets. In addition, privacy concerns regarding user data on digital library platforms must be addressed to maintain public trust.

Technological limitations also affect the digitization process. Certain ancient materials, such as palm leaf manuscripts and fragile parchments, may not withstand scanning procedures. Some artifacts are too fragile to be exposed to light or physical handling, challenging digital preservation (Crisnapati et al., 2019). Advanced techniques such as multispectral imaging and 3D scanning offer potential solutions, but their high cost and complexity limit their widespread adoption in Indonesian libraries.

In addition, digitization diminishes the physical experience of cultural heritage. Interacting with historical materials in person provides a sensory connection that digital formats cannot replicate. The tactile experience of turning the pages of an ancient manuscript or observing the intricate carvings on a temple relief has cultural value. Johnston (2020) argues that although digital archives facilitate access, they cannot replace the authenticity of direct interaction with heritage objects. To address this issue, libraries should integrate virtual reality (VR) and augmented reality (AR) technologies to enhance digital interaction with cultural artifacts.

IV. Technical Challenges in Digital Preservation

Despite its benefits, the study found that digitalization in Indonesia faces several technical barriers. High costs, infrastructure limitations, and a shortage of skilled professionals remain significant challenges. Many libraries and cultural institutions, particularly in rural areas, lack the technological infrastructure to sustain digital archives (P. X. Zhou, 2021). For instance, the digitalization project of the Borobudur Temple reliefs has faced difficulties in funding and technical support, resulting in slow progress in converting historical engravings into high-resolution digital formats (Regina et al., 2023).

Additionally, maintaining digital archives requires ongoing efforts in data storage and system updates. Libraries must ensure that digital files remain readable despite rapid technological advancements. This is particularly problematic in Indonesia, where government funding for digital initiatives fluctuates, impacting long-term sustainability (Nikonova & Biryukova, 2017). In comparison, developed countries have implemented robust strategies for long-term digital preservation, such as the UK's Digital Preservation Coalition, which ensures data continuity despite technological changes (Njonge, 2023).

V. Legal and Regulatory Frameworks for Digital Archives

Another significant finding of this study concerns digitalization's legal and regulatory challenges. Copyright issues, intellectual property rights, and open-access policies remain unclear in Indonesia. Many libraries struggle with licensing agreements when digitizing collections, as copyright laws do not fully accommodate digital repositories (Adetunji & Okuonghae, 2022). The Indonesian Copyright Act No. 28 of 2014 provides legal protection for cultural works but does not comprehensively address the digitalization of heritage materials (Rizkia et al., 2024).

A well-known case highlighting the legal complexities of digital archives is the digitization of *Babad Tanah Jawi*, a historic Javanese chronicle. The manuscript's digital version was made available online, leading to disputes regarding intellectual property ownership. Scholars debated whether the digital format should be freely accessible or restricted due to its cultural significance (Lukman et al., 2020). This case demonstrates the need for Indonesia to adopt more transparent policies similar to the European Union's Directive on Copyright in the Digital Single Market, which explicitly addresses digital reproduction rights for educational and cultural purposes (Isa et al., 2018).

The Indonesian government must refine its copyright and intellectual property policies to support libraries and institutions, ensuring that digital archives remain legally protected while promoting accessibility for educational and research purposes.

VI. Social and Cultural Impacts of Digitalization

The study also highlights how digital preservation influences social and cultural dynamics. Digitalization has helped Indonesians reconnect with their heritage, increasing awareness and appreciation of local traditions. Platforms like Google Arts & Culture have collaborated with Indonesian museums to create virtual exhibits featuring traditional crafts, artifacts, and historical events (Lukman et al., 2020). This initiative has allowed Indonesians to engage with their cultural history from remote locations, fostering greater cultural appreciation among younger generations.

However, digitalization has also changed how people interact with cultural heritage. While digital platforms provide convenient access to historical archives, they may reduce the emphasis on physical interactions with artifacts. The shift from visiting museums to engaging with digital exhibits raises concerns about losing tangible heritage experiences (Cramer & Kott, 2010). For example, some cultural experts argue that viewing batik patterns in digital format cannot replace the hands-on experience of learning traditional dyeing techniques from artisans.

Some cultural institutions have adopted hybrid approaches, combining physical and digital experiences. The Museum Nasional Indonesia offers augmented reality (AR) tours, allowing visitors to physically and digitally explore historical artifacts

(Crisnapati et al., 2019). These interactive features enhance cultural education while maintaining a connection to tangible heritage.

Recommendations for Sustainable Digital Preservation

Given the study's findings, several recommendations can be made to improve digital preservation efforts in Indonesia:

1. **Strengthening Digital Infrastructure:** Increased government investment is needed to provide libraries with adequate servers, cloud storage, and security measures to maintain digital archives.
2. **Enhancing Technical Expertise:** Universities and institutions should offer specialized training in digital archiving to equip librarians and cultural experts with the necessary skills.
3. **Developing Clear Legal Frameworks:** Copyright laws should be updated to address digital repositories, ensuring a balance between legal protection and public accessibility.
4. **Encouraging Public Participation:** Libraries and museums should create interactive digital exhibits that allow users to contribute narratives, memories, and interpretations of cultural heritage.
5. **Integrating Digital and Physical Experiences:** Museums should adopt hybrid models that blend digital technologies with hands-on cultural activities to preserve the authenticity of heritage experiences.

Conclusion

This study underscores the vital role of digital preservation in safeguarding Indonesia's cultural heritage, focusing on the benefits, challenges, and implications of digitalization within libraries. The findings show that while digital preservation enhances accessibility, efficiency, and the longevity of historical materials, it also presents significant technical, legal, and social challenges. To successfully implement digital preservation strategies, libraries need robust infrastructure, clear legal frameworks, and well-trained professionals. Adopting hybrid approaches that blend digital and physical interactions can boost public engagement with cultural heritage. Collaboration among libraries, policymakers, and cultural institutions is essential to ensure sustainable practices. Further research should investigate how emerging technologies, such as artificial intelligence and blockchain, can address these challenges and strengthen digital preservation efforts, particularly in the context of evolving cultural heritage management needs in the digital age.

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Author Contributions

Samuel Ega Oktavianus contributed to conceptualizing the research framework, data collection, drafting the manuscript, literature review, organizing references, and finalizing the manuscript for submission. Efraim Samba Alfarado was responsible for data analysis, results interpretation, and methodology refining.

Declaration of Conflicting Statements

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